FIITJEE

ALL INDIA TEST SERIES

FULL TEST – I

JEE (Main)-2025

TEST DATE: 26-12-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. A

Sol. y_{cm} of solid hemisphere = 3R/8

y_{cm} of solid cone = R/4

2. A

Sol. $-\frac{1}{F}=P=2P_{l_1}+2P_{l_2}+P_m$ ...(1)

$$P_{l_1}=\frac{1}{f_1}=(\mu-1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$$

$$P_{l_1}=[(1.5-1]\left[-\frac{1}{10}-\frac{1}{15}\right]=-\frac{1}{12}$$ ...(2)

$$P_{l_2}=\frac{1}{f_2}=(\mu-1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$$

$$P_{l_2}=\left(\frac{4}{3}-1\right)\left[\frac{2}{15}\right]=\frac{2}{45}$$ ...(3)

$$P_m=-\frac{1}{f}=+\frac{2}{15}$$ ...(4)

$$-\frac{1}{F}=P=2\left[-\frac{1}{12}+\frac{2}{45}\right]+\frac{2}{15}=-\frac{1}{6}+\frac{4}{45}+\frac{2}{15}=\frac{1}{18}$$

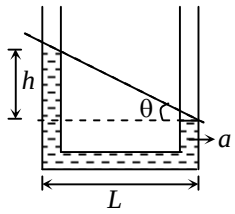
$F=-18\text{cm}$. Focus is negative means system will behave as concave mirror.

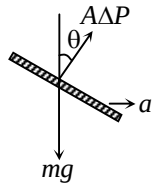
3. D

Sol. Newton's equations are :

$A\Delta P\sin\theta = ma$ …(i)

And $A\Delta P\cos\theta = mg$ …(ii)





By (i) and (ii)

$\tan\theta = \frac{a}{g} = \frac{h}{L}$

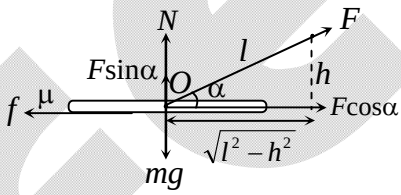
or $h = \frac{aL}{g}$

4. A

Sol. $N = mg - F\sin\alpha$

$F\cos\alpha = f = \mu N$

$F\cos\alpha = \mu(mg - F\sin\alpha)$



$\mu = \frac{F\cos\alpha}{mg - F\sin\alpha} = \frac{F \times \frac{\sqrt{l^2 - h^2}}{l}}{mg - F \times \frac{h}{l}}$

$\mu = \frac{F\sqrt{l^2 - h^2}}{mgl - Fh}$

5. C

Sol. Time taken her oscillation her any tunnel through earth is $\omega = \sqrt{\frac{g}{R}}$ consider earth tunnel and add total displacement.

$A \to S - \frac{\pi}{3}$

$S \to P \ -\frac{\pi}{2}$

$P \to C \ -\frac{\pi}{2}$

Total phase travelled $= \frac{4\pi}{3}$

$T = \frac{4\pi}{3\omega} = \frac{4\pi}{3}\sqrt{\frac{R}{g}}$

6. C

Sol. The net torque produced by the spring when the disc is rotated through an angle θ is

$\tau_C = (kx)R = (I_C + mr^2)\alpha$

Where $x = R\theta$ and $I_c = MR^2$

or $\alpha = \frac{kR^2}{\left(\frac{MR^2}{2} + mr^2\right)}\theta$ or $\omega = \sqrt{\frac{2kR^2}{MR^2 + 2mr^2}}$

7. D

Sol. $\begin{bmatrix}(33+a)(100)=(67+b)(50)\\(67+a)(50)=(33+b)(100)\end{bmatrix}$

$\begin{bmatrix}66+2a=67+b\\67+a=66+2b\end{bmatrix}\times 2$

$65+3a=68$

$a=1$ and $b=1$

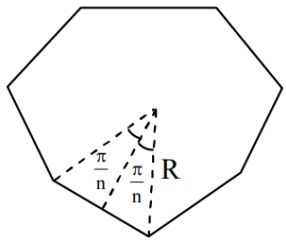
8. B

Sol. $U=-\vec{M}\cdot\vec{B}$

U is max when $\theta = 180°$

9. C

Sol.



$r = R\cos H/n$

B due to one side $= \frac{\mu_0 i}{4\pi r}\left[\sin\frac{\pi}{n}+\sin\frac{\pi}{n}\right]$

$= \frac{\mu_0 i}{2\pi r}\sin\frac{\pi}{n}$

$= \frac{\mu_0 i}{2\pi r}\frac{\sin\pi/n}{R\cos\pi/n}$

$= \frac{\mu_0 i}{2\pi r}\tan\frac{\pi}{n}$

$B_{net} = \frac{n\mu_0 i}{2\pi R}\tan\frac{\pi}{n}$

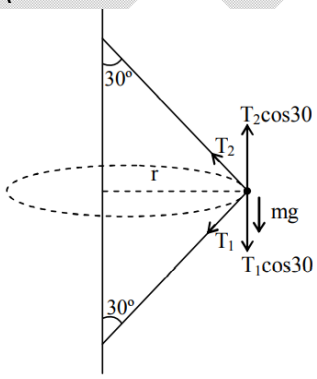
$= \frac{\mu_0 i\tan\pi/n}{2R\frac{\pi}{n}}$

10. D

Sol. Conceptual

11. A

Sol.



r = 0.4 sin 30
= 0.2 m

$$(T_1 + T_2)\sin 30 = \frac{200}{100}10^2 \times 0.2$$

$$\frac{T_1 + T_2}{2} = \frac{0.4}{10}\omega^2$$

$$T_1 \cos 30 + mg = T_2 \cos 30$$

$$T_1 \frac{\sqrt{3}}{2} + \frac{200}{100}10 = T_2 \frac{\sqrt{3}}{2}$$

$$\sqrt{3}\frac{(T_2 - T_1)}{2} = 2$$

$$T_2 - T_1 = \frac{4}{\sqrt{3}}$$

T_1 = 4N given

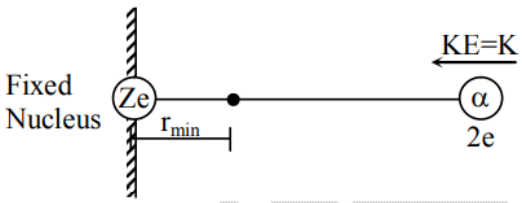
$$T_2 = 4 + \frac{4}{\sqrt{3}} = 6.26 \text{ N}$$

$$\frac{6.26 + 4}{2} = \frac{0.4}{10}\omega^2$$

$$\omega^2 = 128.25 = 11.32 \text{ rad/s}$$

12. C

Sol.



Mechanical energy conservation

$$0 + K = 0 + \frac{Ze \times 2e}{4\pi \in_0 r_{min}}$$

$$r_{min} = \frac{Ze^2 \times 2}{4\pi \in_0 K}$$

$r_{min} \propto Z$

So when Z become 2 Z r_{min} also get double so new r_{min} = 2b

13. A

Sol. $V_1 R_1 = V_2 R_2$

14. B

Sol. Conceptual

15. C

Sol. Conceptual

16. D
Sol. Conceptual

17. C

Sol. Speed of block at the bottom of board = $\sqrt{2gh}$

Applying conservation of linear momentum in horizontal direction,

$m\sqrt{2gh}\cos\alpha = (M+m)v$

$v = \dfrac{m\sqrt{2gh}\cos\alpha}{M+m}$

18. A
Sol. Conceptual

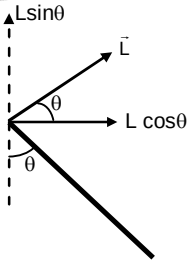
19. C
Sol. Conceptual

20. A
Sol. Conceptual

SECTION – B

21. 6

Sol. $\left|\dfrac{d\vec{L}}{dt}\right| = L\cos\theta\omega$

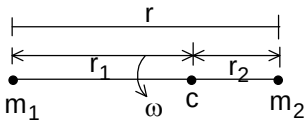


$= \dfrac{m\ell^2}{3}\omega\sin\theta\cos\theta\omega$

$= \dfrac{m\ell^2}{3}\omega^2\sin\theta\cos\theta$

22. 7

Sol. Given that $m_1 - m_2 = 6$ unit …….(i)



equation of motion

$\dfrac{Gm_1m_2}{r^2} = m_1\omega^2 r_1 = m_1\omega^2\dfrac{m_2 r}{m_1+m_2}$

$\therefore m_1 + m_2 = \dfrac{\omega^2 r^3}{G} = 8 \text{ unit } .....(ii)$

by (i) & (ii)

$\dfrac{m_1}{m_2} = 7$

23. 2
Sol. Conceptual

24. 4

Sol. Thermal resistance of AC $R_{AC} = \dfrac{L}{KA} = \dfrac{0.1}{336\times1\times10^{-4}}$

Thermal resistance of BC $R_{BC} = \dfrac{0.2}{336\times10^{-4}} = 2R$

Tc

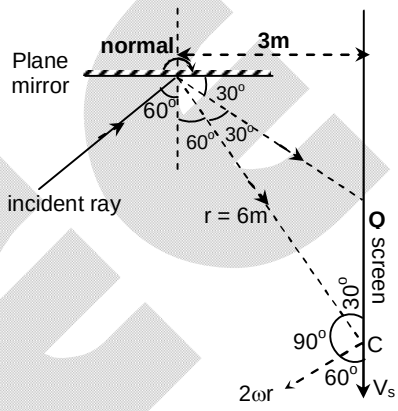
$H_1 = \frac{20}{R}$; $H_2 = \frac{40}{2R} = \frac{20}{R}$

$H = H_1 + H_2 = \frac{40}{R} = 13.44\,W.$

Rate $= \frac{H}{Li} = \frac{13.44/4.2}{80}$ g/s = 40 mg/sec.

25. 2

Sol. When incident ray is fixed the angular velocity of reflected ray becomes twice in the same sense as that of reflected ray. After t = 15 sec, the mirror rotates 15° then reflected ray rotates 30° in the same sense.



In ΔABC, $\cos 60° = \frac{3}{r}$

Where $\omega = \frac{\pi}{180} rad/sec . r = 6m$

Then, there are two components of velocity of spot on the screen.

(a) radial component which increase the length of r i.e.,

$V_s \sin 60° = \frac{dr}{dt}$

(b) perpendicular component

$V_s \cos 60° \quad \Rightarrow V_s \times \frac{1}{2} = 2 \times \frac{\pi}{180} \times 6 \Rightarrow V_s = \left(\frac{2\pi}{15} m/s\right)$

Chemistry

PART – B

SECTION – A

26. C
Sol. Hydrolysis of sucrose brings about a change in the sign of rotation from dextro to laevo.

27. C
Sol. Factual

28. B
Sol. $\left[Fe(CN)_6\right]^{3-}, \left[Mn(CN)_6\right]^{3-}.\left[CoF_6\right]^{3-}, \left[MnBr_4\right]^{2-}$
have 1, 2, 4 & 5 unpaired electron respectively.

29. B
Sol. Factual

30. B
Sol. Factual

31. A
Sol. Tetrahedral complex has intense color.

32. A
Sol. Eu & Gd have half-filled f-orbitals (factual)

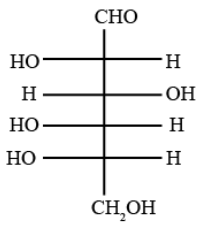
33. C
Sol.
- (2, 4-DNP) test given by aldehyde 1 ketones
- Tollen test
- Canizaro
- Must be at 1, 2 position

34. A
Sol. $pH = pK_a + \dfrac{\log\left[CH_3COO^-\right]}{CH_3COOH}$
C_p is temperature dependent

35. B
Sol. The entropy change for reaction in given by
$\Delta S^\circ_{reactant} = \Delta S^\circ_{product} - \Delta S^\circ_{reactant}$

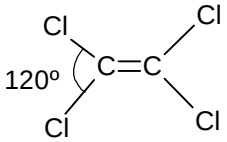
36. C
Sol. Higher order reaction are rare due to low probability of simultaneous collision of all the reacting species.

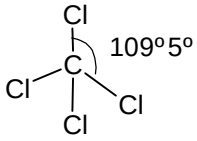
37. A
Sol. The structure of L-Glucose is



38. C
Sol. De pression in free ring point
$\Delta T_f = ik_{fm}$

39. A
Sol. Tetrachloroethene & tetrachloromethane





40. B
Sol. i = 2
π = iCRT

41. B
Sol. Most polar and weakest bond will break first.

42. D
Sol. VSEPR

43. A
Sol.
$$\begin{bmatrix} \Delta G^o = -nFE^\circ_{cell} \\ \Delta S^o = nF\left(\frac{\delta E^\circ_{cell}}{\delta T}\right) \\ \Delta H = \Delta G^o + T\Delta S^o \end{bmatrix}$$

44. B
Sol. $[R] = \frac{K_2}{K_1 + K_2}[P]\left(1 - e^{-(k_1+k_2)t}\right)$

45. B
Sol. % ionic character = $\frac{\text{observed dipole moment}}{\text{calculated dipole momet}} \times 100$

SECTION – B

46. 0
Sol. No. of radical nodes = n – l –1
= 4 –3 –1
= 0

47. 2
Sol. $\frac{[B]}{[C]} = \frac{k_1}{k_2} = \frac{2k_2}{k_2} = 2$

48. 5
Sol. $K_{final}\ P(g) \rightleftharpoons D(g) = K_{C_1} \times K_{C_2} \times K_{C_3}$
$= 10 \times 2 \times 0.01 = \frac{20}{100} \times \frac{2}{10}$
$= 1/5$

$D \rightleftharpoons P(g)$

$k_f^1 = \frac{1}{k_{final}} = \frac{1}{1/5} = 5$

49. 4

Sol. $E_n = -\frac{13.5}{n^2} Z^2$

50. 3

Sol. Paramagnetic species are those which have unpaired of electron present in then NO, O_2 & B_2 have unpaired electron

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Mathematics

PART – C

SECTION – A

51. C

Sol. In expression $\left(ax^2+\frac{1}{bx}\right)^{11}$, we have

$T_{r+1}={}^{11}C_r\left(\frac{a^{11-r}}{b^r}\right)(x)^{22-3r}$ for x^7

$\Rightarrow 22-3r=7 \Rightarrow r=5$

Also in the expansion of $\left(ax-\frac{1}{bx^2}\right)^{11}$

$t_{r+1}=(-1)^r\ {}^{11}C_r.\frac{a^{11-r}}{b^r}.x^{11-3r}$ for x^{-7}

$11-3r=-7 \Rightarrow r=6$

Comparing, $T_6=t_7 \Rightarrow ab=1$

Now $\frac{a^2+b^2}{2}\geq|ab| \Rightarrow a^2+b^2\geq 2$

52. D

Sol. Equation of common chord is $15\alpha x+(\beta+\gamma)y+(\alpha+1)=0$

Comparing with $15x+\delta y-\alpha=0$ passes through A and B for

$\therefore$ No real solution for α.

53. C

Sol. $\arg\left(z(1+\bar{z})\right)+\arg\left(\frac{z\bar{z}}{z(1-\bar{z})}\right)=0$

$\Rightarrow \arg z+\arg(1+\bar{z})+\arg\bar{z}-\arg(1-\bar{z})=0$

$\Rightarrow \arg\left(\frac{1+\bar{z}}{1-\bar{z}}\right)=0$

$\Rightarrow \frac{1+\bar{z}}{1-\bar{z}}=k>0$

$\Rightarrow \bar{z}=\frac{k-1}{k+1}$

$\Rightarrow |\bar{z}|<1$

54. D

Sol. $\left|\sqrt{(x^2-2)^2+(x-3)^2}-\sqrt{(x^2+2)^2+x^2}\right|$

i.e. $\left|\sqrt{(y-2)^2+(x-3)^2}-\sqrt{(y+2)^2+x^2}\right|$

Here $y=x^2$. If P(x, y) be any point on the parabola, then $|PA-PB|\leq AB$

Where A(3, 2), B(0, –2) $\Rightarrow$ AB = $\sqrt{9+16}=5$

55. D

Sol. $f(x)=\frac{x^3+x-2}{x^3-1} \Rightarrow f(x)=\left(\frac{x-1}{x-1}\right)\left(\frac{x^2+x+2}{x^2+x+1}\right)=\frac{x^2+x+2}{x^2+x+1}$, when $x\neq 1$

Hence range is $\left(1,\frac{7}{3}\right]$.

56. D

Sol. Total cases = $2 \times 3 \times 3 \times 2 = 36$

57. B

Sol. $t_n = \cot^{-1}\left(2 + \frac{n(n+1)}{2}\right)$

$= \tan^{-1}\left(\frac{2}{4+n(n+1)}\right) = \tan^{-1}\left(\frac{\frac{1}{2}}{1+\frac{n}{2}\left(\frac{n+1}{2}\right)}\right) = \tan^{-1}\left(\frac{n+1}{2}\right) - \tan^{-1}\frac{n}{2}\right)$

$\Rightarrow S_\infty = \frac{\pi}{2} - \tan^{-1}\frac{1}{2} = \cot^{-1}\frac{1}{2} = \tan^{-1}2$

58. D

Sol. (i) $a \neq b \neq c$ all (distinct) and $c \neq 0$ — Total cases $2\,{}^9C_3$

(ii) $a \neq b$, $c = 0$ $(a = b > c)$ — Total cases ${}^9C_2$

(iii) $a = b$, $c \neq 0$ $(a = b > 0)$ — Total cases ${}^9C_2$

(iv) $a = b$, $c = 0$ $(a = b > 0)$ — Total cases ${}^9C_1$

(v) $a = c \neq 0$, $(a < b > c)$ — Total cases ${}^9C_2$

$2\,{}^9C_3 + 3\,{}^9C_2 + {}^9C_1$

59. C

Sol. If n = even $\Rightarrow$ N is odd

If n = odd N is even for b = 26, 27, …..35 and 25 and N is odd for b = 17, 18, ….24

60. A

Sol. $b + c = 2a$ and $g_1^3 = b^2c$

$g_2^3 = bc^2$

Now $\frac{g_1^3 + g_2^3}{abc} = 2$

61. D

Sol. $\text{Log}_6(abc) = 6$

$\Rightarrow (abc) = 6^6$

Let $a = \frac{b}{r}$ and $c = br$

$\Rightarrow b = 36$ and $a = \frac{36}{r} \Rightarrow r = 2, 3, 4, 6, 9, 12, 18$

Also, $36\left(1 - \frac{1}{r}\right)$ is a perfect cube.

$\Rightarrow 36\left(1 - \frac{1}{r}\right)$ is a perfect cube for r = 4.

$\Rightarrow a + b + c = 36 + 9 + 144 = 189.$

62. A

Sol. $1 + a = \frac{4 - 3z}{z^2 - 3z + 4}$

Where $z = \frac{x^2}{1+x^2}$; $0 \leq z < 1$

Let $f(z)=\frac{4-3z}{z^2-3z+4}$, then $f'(z)=\frac{z(3z-8)}{(z^2-3z+4)^2}$

f(z) is decreasing in [0, 1), $\Rightarrow \frac{1}{2}<1+a\le 1;\ -\frac{1}{2}<a\le 0.$

63. B

Sol. XY is parallel to BC

64. B

Sol. $\cos\frac{\pi}{15}\cos\frac{2\pi}{15}\cos\frac{4\pi}{15}\cos\frac{7\pi}{15}=\frac{1}{16}$

$\cos\frac{3\pi}{15}\cos\frac{6\pi}{15}=\frac{1}{4}$

$\cos\frac{5\pi}{15}=\frac{1}{2}$

$\Rightarrow$ required value is $\frac{1}{128}$

65. A

Sol. $\overrightarrow{AB}$ can be in any direction but not in the plane of 2i + 3j + 4k and 3i + 7j + 6k since lines are skew.

66. A

Sol. When $[\alpha] = 0, 1, -1$.

67. A

Sol. $f(x)=\frac{4}{\pi}(\pi-\cot^{-1}(-x))-\frac{\pi}{4\cot^{-1}(-x)}$

$=4-\left(\frac{4\cot^{-1}(-x)}{\pi}+\frac{\pi}{4\cot^{-1}(-x)}\right)\le 4-2=2$

Equality when $\cot^{-1}(-x)=\frac{\pi}{4}$

$\Rightarrow x=-1$

68. B

Sol. Area = $37\times 3-\int_0^3(x^3+3x+1)dx$

69. B

Sol. $S_1P=S_2P\Rightarrow a-e\alpha=E\alpha-\frac{a}{2}.$ Also, $\alpha=\frac{ae+\frac{a}{2}E}{2}$

Eliminating α we get, $E^2+3eE+(2e^2-6)=0$

Now, as $e=\sqrt{1-\frac{b^2}{a^2}}=\frac{1}{\sqrt{7}}, E=\frac{5}{\sqrt{7}}\Rightarrow\frac{a}{c}=\frac{1}{\sqrt{E^2-1}}=\frac{\sqrt{7}}{\sqrt{18}}$

70. B

Sol. $\sum x = 40 \times 200 - 50 + 40 = 7990$

Corrected $\bar{x} = \dfrac{7990}{200} = 39.95$

Incorrect $\sum x^2 = n\left[\sigma^2 + \bar{x}^2\right] = 200\left[15^2 + 40^2\right] = 365000$

Correct $\sum x^2 = 36410 - 50^2 + 40^2 = 364100$

Correct $\sigma = \sqrt{\dfrac{364100}{200} - \left(39.95\right)^2} = 14.98$

SECTION – B

71. 1

Sol. $\dfrac{2f(x).f'(x)}{\sqrt{1-(f(x))^4}} - 2x \geq 0 \Rightarrow \dfrac{d}{dx}(\sin^{-1}(f(x))^2 - x^2) \geq 0$

Let $g(x) = \sin^{-1}((f(x))^2) - x^2$ is a non-decreasing function.

$\Rightarrow \lim\limits_{x\to x_1} g(x) \leq \lim\limits_{x\to x_2} g(x) \Rightarrow \dfrac{\pi}{2} - x_1^2 \leq \dfrac{\pi}{6} - x_2^2$

$\Rightarrow x_1^2 - x_2^2 \geq \dfrac{\pi}{3} \Rightarrow \left[x_1^2 - x_2^2\right] \geq 1.$

72. 1

Sol. $\lim\limits_{x\to 1} \sec^{-1}\left(\dfrac{\lambda^2}{\ln x} - \dfrac{\lambda^2}{x-1}\right)$

Let $x - 1 = t \Rightarrow x = t + 1 \Rightarrow$ as $x \to 1, t \to 0$

$= \lim\limits_{t\to 0} \sec^{-1}\left(\dfrac{\lambda^2}{\ln(t+1)} - \dfrac{\lambda^2}{t}\right) = \lim\limits_{t\to 0} \sec^{-1}\left(\dfrac{\lambda^2}{t}\left(\dfrac{\lambda^2}{\ln(t+1)} - 1\right)\right)$

$= \lim\limits_{t\to 0} \sec^{-1}\left(\dfrac{\lambda^2}{2}\right) \qquad \Rightarrow \dfrac{\lambda^2}{2} \geq 1 \,|\, \lambda \,| \geq \sqrt{2}$

Hence minimum value of $[|\lambda|] = 1.$

73. 0

Sol. Since $\vec{x}.\vec{a} = \vec{x}.\vec{b} = \vec{x}.\vec{c} = 0$

$\Rightarrow (\vec{a}, \vec{b}, \vec{c}$ are coplanar) $\vec{x}$ is perpendicular to $\vec{a}, \vec{b}, \vec{c}$

$\therefore \vec{a}.(\vec{b}\times\vec{c}) = 0$

74. 2

Sol. $x^2y^2\left(\dfrac{dy}{y^2} - \dfrac{dx}{x^2}\right) + x^2y^2\left(\dfrac{1}{y} - \dfrac{1}{x}\right)dy = 0$

$\Rightarrow d\left(\dfrac{1}{x} - \dfrac{1}{y}\right) + y\left(\dfrac{1}{y} - \dfrac{1}{x}\right)dy = 0 \Rightarrow \dfrac{d\left(\dfrac{1}{x} - \dfrac{1}{y}\right)}{\dfrac{1}{x} - \dfrac{1}{y}} = ydy$

$\Rightarrow \ln\left|\dfrac{1}{x} - \dfrac{1}{y}\right| = \dfrac{y^2}{2} + c \Rightarrow k = 2.$

75. 3

Sol. $f(x) = \sin x + \int_{-\pi/2}^{\pi/2} (\sin x + t\cos x)\left(-\frac{1}{k}\sin t - \frac{2}{k}\cos t\right)dt$

$= \sin x + I_1 + I_2 + I_3 + I_4$

Where $I_1 = -\frac{\sin x}{k}\int_{-\pi/2}^{\pi/2} \sin t\,dt = 0$

$I_2 = -\frac{2\sin x}{k}\int_{-\pi/2}^{\pi/2} \cos t\,dt = -\frac{4\sin x}{k}$

$I_3 = -\frac{\cos x}{k}\int_{-\pi/2}^{\pi/2} t\sin t\,dt = -\frac{2\cos x}{k}$

$I_4 = \frac{4\cos x}{k}\int_{-\pi/2}^{\pi/2} t\cos t\,dt = 0 \qquad (\because t\cos t \text{ is odd})$

$\Rightarrow -\frac{1}{k}\sin x - \frac{2}{k}\cos x = \sin x - \frac{4\sin x}{k} - \frac{2\cos x}{k}$

$\Rightarrow -\frac{1}{k} = 1 - \frac{4}{k} \Rightarrow k = 3.$